

Marco Aurelio Menezes

Data Engineer

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SUMMARY

Senior Data Engineer – DE with over 8 years of experience in Python programming language, 4 years of work experience, designing and implementing data pipelines related with Big Data workloads, using Python, Spark, Hadoop Ecosystem, and SQL for validations. Over 2 years migrating data pipelines and systems from on-premises to Cloud - Azure + Databricks. Since this year responsible for migrating Accounting streaming data pipelines to AWS. Background in Control and Automation Engineering studies. Known for the expertise in Python, Spark and big data fundamentals with a passion about sharing knowledge. Proven ability to integrate DE concepts with new tools and frameworks, as technology evolves, always balancing purpose, performance, costs and other trade-offs.

SKILLS

Proficient: Python, SQL, Spark, Linux, Cloud Migration, Purpose-driven, Communication and Decision-Making.

Intermediate: Java, Scala, PostgreSQL and Aurora, Scylla, DynamoDB, and Redis databases, Apache Kafka, Airflow and Iceberg, AWS & Azure DE related services, Databricks, DE team Leadership and Development life cycles using Agile and CI/CD methodologies and tools such as Jira, Git, Jenkins and Service Now.

Beginner: Apache Flink and Machine Learning algorithms.

PROFESSIONAL EXPERIENCE

Senior Data Engineer

F1rst Digital Services – Santander Brazil, São Carlos – São Paulo, BR

02/2024 to Present

Core Responsibilities:

- Responsible for conducting the DE accounting team through the Data Transformation strategic movement;
- The Data Transformation consists of migrating batch and streaming data pipelines from on-premises to Cloud, adapting them to a Data Mesh and Medallion Architecture design;
- Migration pipelines from on-premises to the Azure using Databricks, Data Factory and ADLS gen2.
- Migration of streaming data pipelines from DB2, Kafka, Spark Streaming, ScyllaDB and OpenShift to AWS using the resources DB2, Kafka, Amazon MSK and MSK connectors, Aurora and EKS.
- Design of new daily batch data pipelines using Azure and Databricks to deliver Bronze and Silver Accounting Delta tables to our Data Domain, named Cash Management;

Key Technologies and Tools: Spark, Hadoop, Hive, Databricks, Azure Data Factory and Data Lake Storage, and AWS services such as Amazon MSK, Aurora, EKS, S3 and CloudWatch.

Mid-level Data Engineer

F1rst Digital Services – Santander Brazil, São Carlos – São Paulo, BR

07/2022 to 02/2024

Core Responsibilities:

- Responsible for conducting the Cloud migration for the Operational Discontinuity Plan system;
- The system must increase durability of Analytical big data tables by copying partition of daily and monthly Hive tables to a secondary Hadoop cluster located 100 Km far away;
- As the sources were migrated to Azure, the system evolved to copy Delta Tables located in different data domains data lakes, and stored using Local Redundancy using Cool / Hot tiers configuration to a Zone Redundancy Storage using the archive tier, therefore improving durability and reducing storage costs;
- Due to the nature of the system, with high volume and sensitive data, big challenges related to governance, compliance, and security were overcome, demanding assertive communication and purpose-driven skills;

Key Technologies and Tools: Python, Spark, Shell-Script, Hadoop Ecosystem, Databricks, and Azure Cloud;

Junior Data Engineer

F1rst Digital Services – Santander Brazil, São Carlos – São Paulo, BR

05/2021 to 07/2022

Core Responsibilities:

- Design, implementation and maintenance of data pipelines related to big data workloads for reporting;
- Data delivery for regulatory purposes ensuring bank financial healthy and law compliance.
- In June 2022 certified as Advanced Data Engineer by Santander Brazil, ranking among the top 10 data engineers in the institution, after passing a test and presenting a technical case to data experts.

Key Technologies and Tools: Python, Spark, SQL, Hadoop, Hive, Hue, Shell-Script

Junior Data Engineer

Trie Engineering, Belo Horizonte, BR – Remote

10/2020 to 05/2021

Core Responsibilities:

- Maintenance of low-volume data pipelines fed by Google Spreadsheets and field data;
- Provide dashboard data feeds to support decision-making for stakeholders in the Cement Industry.

Key Technologies and Tools: Python, Pandas and Numpy.

Scholarship recipient

Fire labs, Federal University of Ouro Preto, BR

08/2016 to 08/2017

- Development of embedded software to control a gas oven equipment using data from temperature and gas flow sensors to control a gas valve and a AC motor to inject air in the system to follow a curve temperature.

Key Technologies and Tools: C, Arduino, Control Systems design.

Summer Internship

Analógica Instrumentation and Control, Belo Horizonte, BR

01/2015 to 02/2015

- Development of an application to extract data from fuel tank sensors and ingest it for downstream apps.
- First work experience with Python.

Key Technologies and Tools: Python and Raspberry PI.

EDUCATION

Bachelor of Control And Automation Engineering

Federal University of Ouro Preto, Minas Gerais, Brazil

08/2018

- **Relevant coursework:** Electronics, embedded systems, control systems design, C and Python.
- **Bachelor Thesis:** Analysis of the Dynamics and Design of Temperature Controllers for Electric Showers.

CERTIFICATES

Santander Brazil Data Master Program – Data Engineer Advanced

Jun/2022

Databricks - Data Engineer Associate

Jun/2023

Microsoft Azure - Data Engineer Associate – DP-203

Ago/2023

AWS - Data Engineer Associate – DEA-C01

Feb/2025

Databricks - Apache Spark Developer

Sep/2024

Microsoft Azure - AZ-900, DP-900, SC-900 and AI-900

2023 and 2024

AWS - Cloud Practitioner

Jul/2024

EXTRACURRICULAR ACTIVITIES

- Exchange Student in Stuttgart, Germany from June to December 2019.
- Social Projects Director of the Association of Federal Student Housing in Ouro Preto from 2014 to 2017.

LANGUAGES

Portuguese: Native

English: Advanced

German: Intermediate